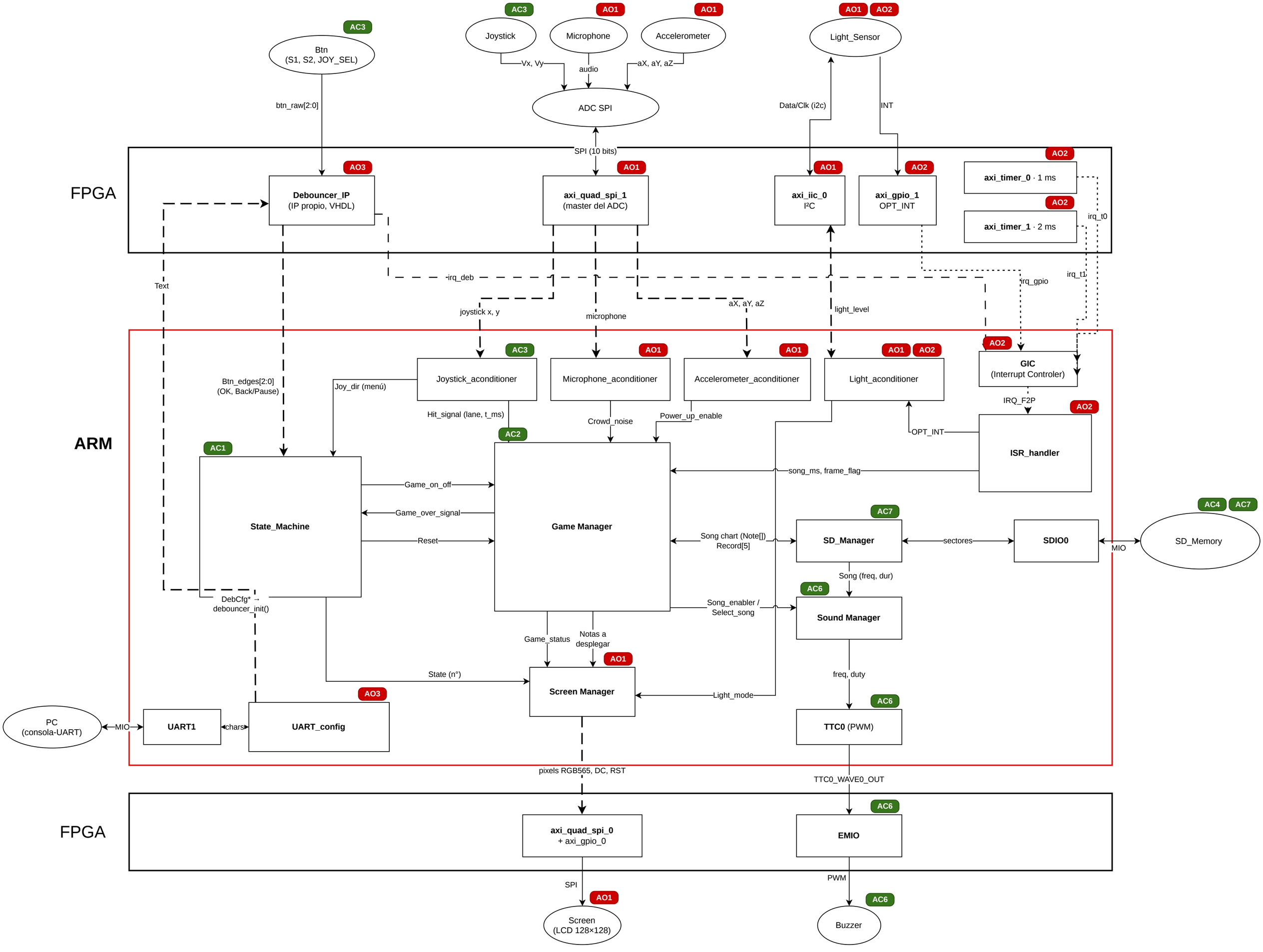

AC3
Btn
(S1, S2, JOY_SEL)
AC3
Joystick
AO1
Microphone
AO1
Accelerometer
AO1
AO2
Light_Sensor
Vx, Vy
audio
aX, aY, aZ
ADC SPI
btn_raw[2:0]
Data/Clk (i2c)
INT
SPI (10 bits)
FPGA
AO3
Debouncer_IP
(IP propio, VHDL)
AO1
axi_quad_spi_1
(master del ADC)
AO1
axi_iic_0
I²C
AO2
axi_gpio_1
OPT_INT
AO2
axi_timer_0 · 1 ms
AO2
axi_timer_1 · 2 ms
irq_t0
irq_t1
irq_gpio
irq_deb
Text
joystick x, y
microphone
aX, aY, aZ
light_level
ARM
AC3
Joystick_aconditioner
AO1
Microphone_aconditioner
AO1
Accelerometer_aconditioner
AO1
AO2
Light_aconditioner
AO2
GIC
(Interrupt Controler)
Btn_edges[2:0]
(OK, Back/Pause)
Joy_dir (menú)
Hit_signal (lane, t_ms)
Crowd_noise
Power_up_enable
IRQ_F2P
AO2
ISR_handler
OPT_INT
AC1
State_Machine
AC2
Game Manager
song_ms, frame_flag
Game_on_off
Game_over_signal
Reset
AC7
SD_Manager
sectores
SDIO0
AC4
AC7
SD_Memory
MIO
Song chart (Note[])
Record[5]
Song (freq, dur)
AC6
Sound Manager
DebCfg* →
debouncer_init()
Song_enabler /
Select_song
Game_status
Notas a
desplegar
AO1
Screen Manager
State (n°)
Light_mode
freq, duty
AC6
TTC0 (PWM)
AO3
UART_config
PC
(consola-UART)
MIO
UART1
chars
pixels RGB565, DC, RST
TTC0_WAVE0_OUT
FPGA
AC6
EMIO
axi_quad_spi_0
+ axi_gpio_0
PWM
SPI
AO1
AC6
Screen
(LCD 128×128)
Buzzer